

UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2024/2025

SEPTEMBER EXAMINATION

UCCN2243 INTERNETWORKING PRINCIPLES AND PRACTICES

WEDNESDAY, 9 OCTOBER 2024

TIME : 9.00 AM – 11.00 AM (2 HOURS)

BACHELOR OF INFORMATION TECHNOLOGY (HONOURS)
COMMUNICATIONS AND NETWORKING
BACHELOR OF INFORMATION TECHNOLOGY (HONOURS)
COMPUTER ENGINEERING
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
INFORMATION SYSTEMS ENGINEERING
BACHELOR OF COMPUTER SCIENCE (HONOURS)

Instruction to Candidates :

This question paper consists of **THREE (3)** questions in **Section A** and **TWO (2)** questions in **Section B**.

Answer **ALL** questions in **Section A** and **ONLY ONE (1)** question in **Section B**.
Each question carries 25 marks.

Should a candidate answer more than **ONE (1)** question in **Section B**, marks will only be awarded for the **FIRST** question in that section in the order the candidate submits the answers.

Candidates are allowed to use calculator.

Answer questions only in the answer booklet provided.

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Section A (Compulsory Questions)

Q1. You are assigned the following network 202.100.148.0/22. Using variable-length subnet masking (VLSM), create subnets as required for the network shown in Figure 1 (assume IP subnet-zero). Ensure that you always utilize the next available network in your IP plan.

(a) **Calculate** the network address, and subnet masks for the following areas. Hint: The number of usable addresses in Area 7 corresponds to the number of gateway interfaces.

(i) Area 1 to Area 3 (7 marks)

(ii) Area 4 to Area 7 (8 marks)

(b) **Determine** the gateway address of Area 3, Area 4 and Area 5. Assume the respective gateway address is the second-last usable address. (5 marks)

(c) There is an error for one of the connection link used in Figure 1. **Identify** the error and **suggest** the proper connection link to be used. (5 marks)

[Total: 25 marks]

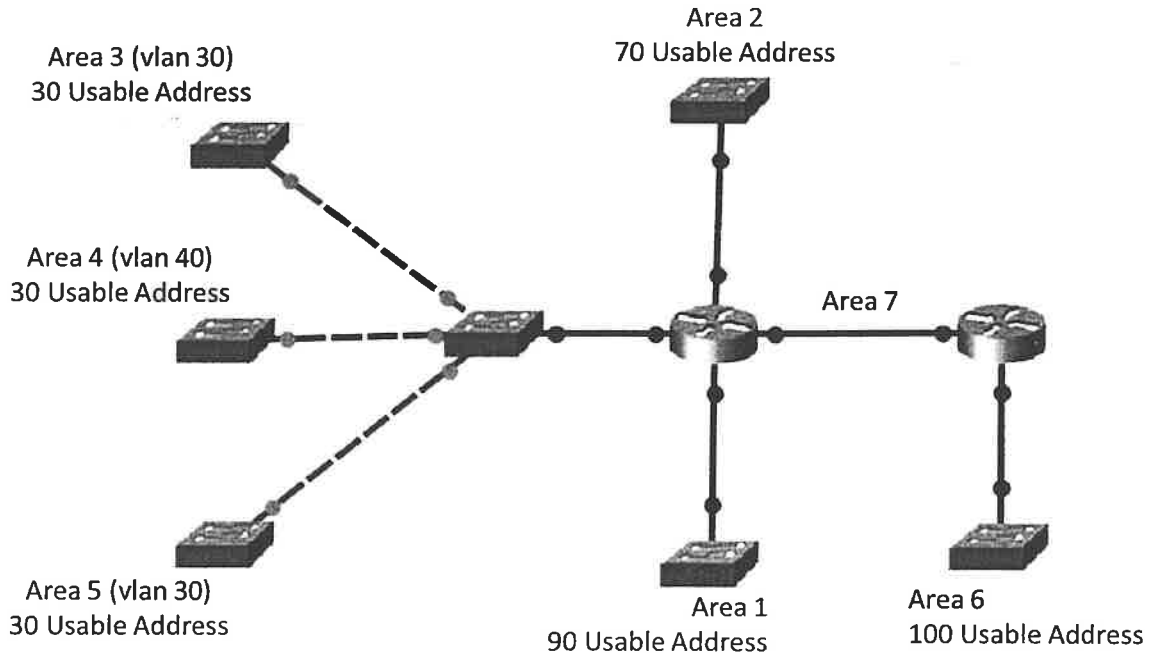


Figure 1

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Q2. Refer to the network in Figure 2. Company AA has a public IP address of 200.1.1.2. Network Address Translation (NAT) is configured on router R1 to enable the HR Department to access the Internet as well as the public to access the Web Server.

- (a) Due to security reasons, only devices from the HR department with the IP address 192.168.1.1 to 192.168.1.15 are allowed to access the Internet using the public IP address. **Show** the relevant Access Control List (ACL) and Port Address Translation configurations for the following interfaces of router R1. Assume the necessary IP addresses have been configured in all the router interfaces. (Hint: You will only need to utilize the information given in Figure 2 for your setup).

(i) Global configuration mode (9 marks)

(ii) Interface S1 and G1 (5 marks)

- (b) **Show** the relevant Port Forwarding configurations of router R1 to enable traffic from the Internet to access the Hypertext Transfer Protocol Secure (HTTPS) web server. (4 marks)

- (c) Supposed Company AA decided to add a File Transfer Protocol (FTP) Server in the IT department. **Discuss** in detail the necessary Port Forwarding configurations to allow traffic from the Internet to access the service using the same public IP address. (7 marks)

[Total: 25 marks]

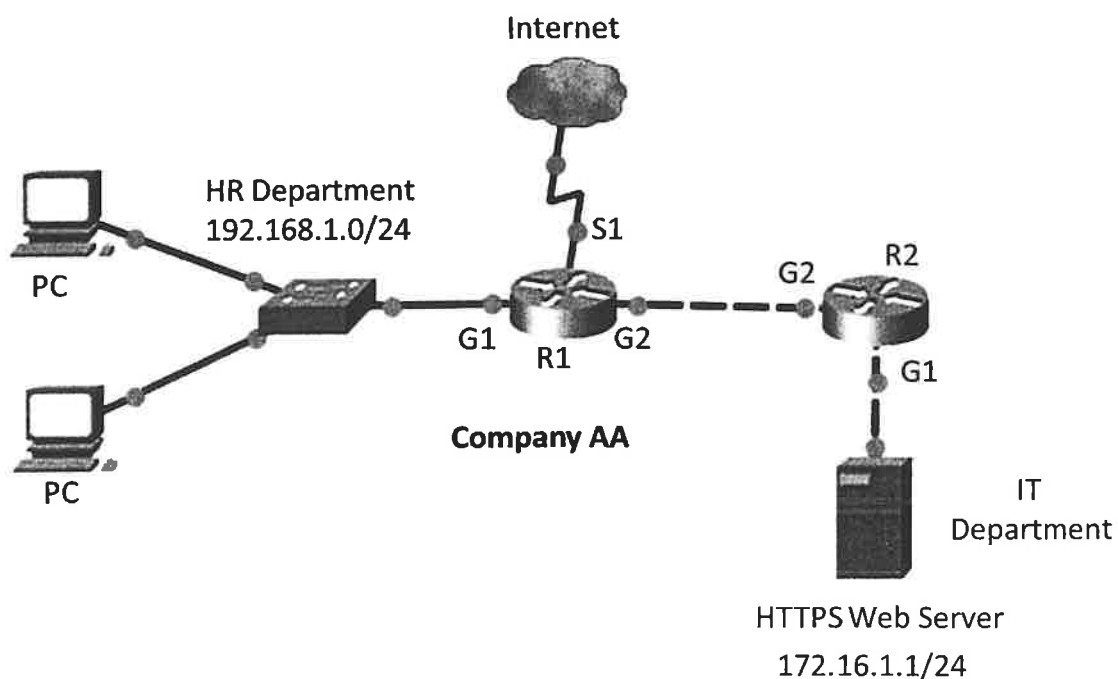


Figure 2

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- Q3. (a) Figure 3 shows an IPv6 network with the relevant PC1 IPv6 address and the network address for subnet 2, subnet 3 and subnet 4.
- Show** all the necessary commands to configure interface Fa0/1 of router R3 with the 1st usable IP address as the gateway for Subnet 1. (5 marks)
 - Determine** IP addresses for interface F0/0 of routers R1, R2 and R3 supposing they are the 1st usable, 2nd usable and 3rd usable address respectively. (3 marks)
 - Referring to the answers in (a)(ii), **show** the necessary commands to configure router R2 with static routes to all networks. You may assume the basic network setup on the router has been done. (6 marks)
 - Determine** the link local address of Server 1 if its media access control (MAC) address is 00-01-43-5D-54-8A. (5 marks)
- (b) Briefly **discuss** THREE(3) techniques used for IPv4-IPv6 transition and coexistence. (6 marks)
- [Total: 25 marks]

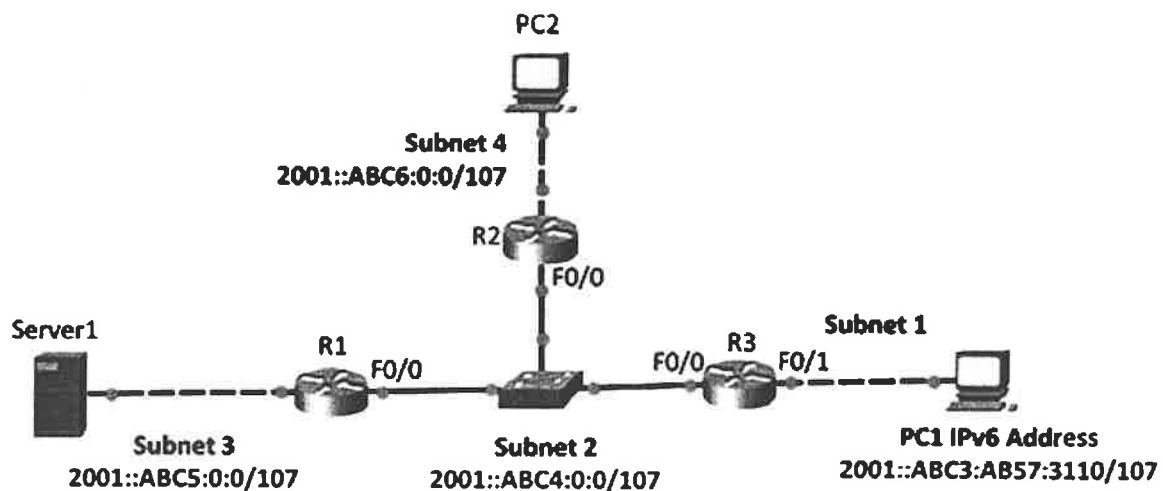


Figure 3

UCCN2243 INTERNETWORKING PRINCIPLES AND PRACTICES**Section B (Choose Any One Question)**

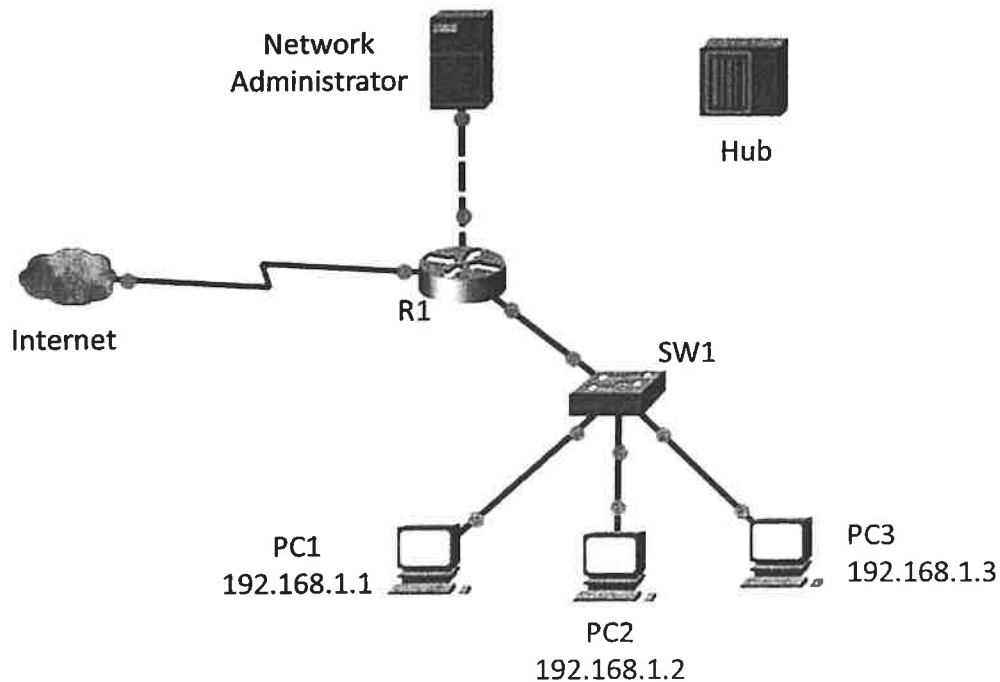
- Q4. (a) A PC has been configured with the following iptables commands as shown in Table 1. **Explain** the actions of these commands on the specified network traffic to the PC. (6 marks)

```
#iptables -A INPUT -p icmp --icmp-type echo-request -m limit --limit 2/second -j ACCEPT

#iptables -A INPUT -p icmp --icmp-type echo-request -m limit --limit 3/minute -j LOG
```

Table 1

- (b) Given the network shown in Figure 4. Explain how PC2 could divert the packets of PC1 to itself by exploiting the ARP protocol. (8 marks)
- (c) **Redraw** the network of Figure 4 and **describe** how the Network Administrator could monitor the Internet traffic of PC1, PC2 using the Hubbing Out technique. (7 marks)
- (d) Continue from Q4.(c), using iptables, **configure** PC1 to drop any packets from PC2. (4 marks)
- [Total: 25 marks]

**Figure 4**

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Q5. (a) Figure 5 shows the data dump (in hexadecimal) at the beginning of an IP packet. Assume all the IP packet headers have the same options. Answer the following questions:

- (i) **Calculate** the size (in bytes) of the initial IP data/payload. (3 marks)
- (ii) Suppose the datagram will be routed to an Ethernet network, **calculate** the total length for each of the fragments. (9 marks)
- (iii) Continue from Q5.a(ii), **state** the number of fragments that are created. (2 marks)
- (iv) Continue from Q5.a(iii), **determine** the fragment offset for each of the fragments. (3 marks)

- (b) **Describe** FOUR (4) characteristics/features of the Transmission Control Protocol (TCP). (8 marks)

[Total: 25 marks]

4A 00 0B B8 10 00 00 00

